

AUTOMATION PORTFOLIO — TECHNICAL RECORD

SOPHON FINANCE SYSTEMS

AI-Driven Finance & Accounting Automation | github.com/sophonfinance-wq/finance-automation-portfolio

CROSS-BORDER TAX SURPLUS / ACB

A deterministic, formula-driven calculation engine that computes Canadian foreign-affiliate surplus pools (exempt / taxable / pre-acquisition capital) and adjusted cost base (ACB) across a multi-tier ownership chain for CRA Form T1134 support, with an independent reconciliation harness and a human-gated assurance loop.

Report No.	SFS-ENG-003	Package	surplus_engine
System	Cross-Border Tax Surplus / ACB	Data	Fictional / seeded
Run	<code>python -m surplus_engine --start 2021 --end 2024 --out out --xlsx</code>	Status	Public — portfolio record

1.0 OVERVIEW AND SCENARIO

A Canadian-parented group holds US operating entities through the seeded four-tier chain Birchwood Op Co (USD, 80% owned) → Cedar Mezz Holdings LLC (USD, 90%) → Maple Fund LP (USD, 100%) → Demo Holdings Inc. (CAD top holdco), and must track, year by year and entity by entity across FY2021–2024, how much of each tier's accumulated earnings is exempt surplus, taxable surplus, or a return of pre-acquisition capital, plus the ACB of each investment. The engine catches errors this work is prone to: for example, the assurance-loop demo injects a fat-fingered 2021 closing exempt-surplus cell (+50,000), an intercompany elevation booked at Maple Fund with no supporting subsidiary distribution (+8,000), and a mis-keyed 2023 mezzanine ACB balance (+15,000), then drives the drifted workpapers back to a clean tie-out. It also surfaces a blended-rate FX trap on Cedar Mezz, where a 35,502.71 contribution at rate 1.3383 and an equal return of capital at 1.3569 net to zero ACB in functional currency yet to (660.35) CAD on a per-layer translation.

2.0 WHAT THE SYSTEM AUTOMATES

- Per-entity, per-fiscal-year computation of standalone surplus (standalone taxable income plus a Reg. 5907(2)-style book-tax adjustment) and its split into exempt vs taxable pool additions by an exempt portion
- The distribution waterfall that consumes pools in statutory order (exempt → taxable → pre-acquisition capital) subject to a 60% exempt-distribution cap on tax-free repatriation
- Surplus elevation up one tier only on an actual subsidiary distribution, at the owner's ownership percentage, so operating income alone never elevates
- ACB tracking that moves only on capital events (contributions increase it, returns of capital decrease it), floors ACB at nil, and surfaces an ITA 40(3)-style deemed gain when a return of capital exceeds basis
- Roll-forward continuity so each entity's closing balances in year N seed its opening balances in year N+1
- Per-layer (lot-level) FX translation of ACB, translating each capital event at its own year's rate and comparing it against the single-rate Summary-layer figure to expose magnitude and sign divergences

- Generation of Markdown per-entity workpapers, a CAD consolidated summary, an FX-layer analysis, a reconciliation report, and an optional openpyxl Excel workbook
- A bounded, human-gated assurance loop that detects the earliest drifted period, re-derives it from source, books field-level journal adjustments, locks the period, and re-verifies until convergence

3.0 OPERATING PROCEDURE

- 1 Generate the fictional four-tier structure, per-entity/per-year facts, and USD→CAD FX table deterministically from a fixed seed (stdlib random, no external data)
- 2 Process fiscal years in ascending order and, within each year, walk entities bottom-up so a subsidiary's distribution is known before its parent elevates it
- 3 Compute standalone surplus, split current-year additions into exempt/taxable pools, and add any elevation received from subsidiaries
- 4 Apply capital contributions to the pre-acquisition pool and ACB, then run the distribution waterfall (exempt→taxable→pre-acq) under the exempt-distribution cap
- 5 Reduce ACB for returns of capital, floor it at nil, and record any deemed capital gain on the excess over basis; commit closing balances as next year's opening
- 6 Attach FX rates and render per-entity workpapers, the CAD consolidated summary, the per-layer FX analysis, and (optionally) the Excel workbook
- 7 Run the reconciliation harness, which independently re-derives every statutory quantity from published intermediates and checks each named identity with expected, actual, and signed delta
- 8 Under --check, fail the process on any identity break so the harness acts as a CI gate; the assurance loop (loop.py) additionally remediates drift period-by-period and returns a PASS/FLAG/FAIL verdict as an exit code

4.0 CONTROLS AND SAFEGUARDS

- Deterministic core: seeded stdlib random (DEFAULT_SEED 20240607) and headless recalculation make every run reproducible; the loop's settled workpapers are byte-identical to a clean engine run
- Independent re-derivation: the reconciliation harness rebuilds the roll-forward from exposed intermediates rather than trusting stored balances, catching any regression that makes internal bookkeeping disagree with its own published figures
- Source-of-truth hierarchy (external workpaper → cited evidence tab → calculation → summary) so every figure traces back to a source and formulas are not hardcoded
- Named-identity tie-outs: 15 structural identities (pool conservation, deemed gain, waterfall integrity, roll-forward continuity, elevation, and the FX layer tie) each carry a signed delta pinpointing the entity, year, and quantity that breaks
- Model-level invariants enforced in code: pool balances never go negative, exempt draws never exceed the cap, ACB is deemed nil rather than negative, and returns of capital cannot exceed the distribution
- Built-in FX tie-out: signed functional-currency ACB layers must sum back to the engine's closing ACB, so the per-layer FX model cannot drift from the engine
- Human gate and separation of concerns: the assurance loop books adjustments only from the engine's own source re-derivation, never invents a number, and FLAGs material adjustments (above a CAD 1,000 materiality threshold) for human review before sign-off

- CI enforcement: --check exits non-zero on any identity break, and the loop verdict maps FAIL to a non-zero exit code

5.0 VERIFICATION AND TESTING

Correctness is proven three ways from the repository itself. A pytest suite reports 1,486 passing tests asserting the load-bearing rules (waterfall ordering, exempt-cap enforcement, distribution-only elevation at ownership %, ACB untouched by operating income, roll-forward continuity, deemed gain on negative ACB, and the per-layer FX cases). The reconciliation harness reports 212 of 212 identity checks passing on the clean 2021–2024 run and fails the process on any break. The assurance-loop demo independently demonstrates detection and remediation, clearing 6 injected breaks across 3 turns and returning a FLAG verdict with an exit code.

6.0 KEY FACTS

Attribute	Value
Ownership chain modelled	4 tiers (Birchwood Op Co → Cedar Mezz → Maple Fund LP → Demo Holdings Inc.)
Named structural identities	15
Reconciliation checks per pass (FY2021–2024)	212 / 212 pass
Test suite	1,486 tests passing
Exempt-distribution cap	60% (0.60)
Assurance-loop demo	3 injected faults, 6 breaks cleared in 3 turns, verdict FLAG

This record documents a self-contained system in the Sophon Finance Systems automation portfolio. All data is fictional and seeded; no employer or client workpaper, entity, methodology, path, or figure is reproduced. Prepared 14 July 2026.